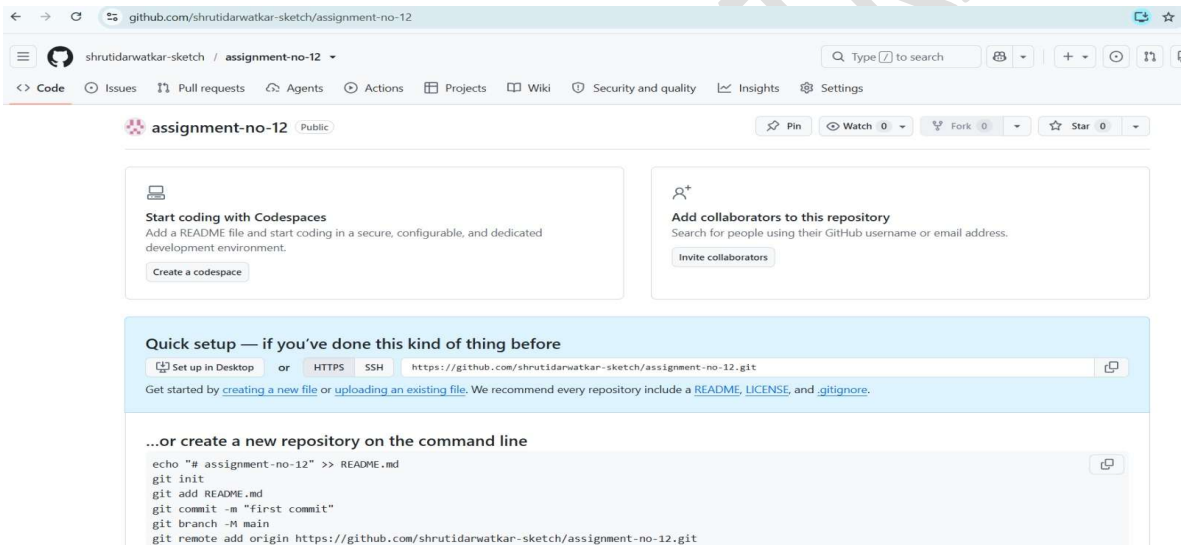


Session 12 (AIML) -Assignment Question

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML

Q1.



```

PS C:\Users\tanve\Desktop\demo project> git --version
git version 2.54.0.windows.1
PS C:\Users\tanve\Desktop\demo project> git init
>>
Reinitialized existing Git repository in C:/Users/tanve/Desktop/demo project/.git/
PS C:\Users\tanve\Desktop\demo project> git add README.md
>>
fatal: pathspec 'README.md' did not match any files
PS C:\Users\tanve\Desktop\demo project> git commit -m "first commit"
On branch main
nothing to commit, working tree clean
PS C:\Users\tanve\Desktop\demo project> echo "# assignment-no-12" >> README.md
PS C:\Users\tanve\Desktop\demo project> git add README.md
PS C:\Users\tanve\Desktop\demo project> git commit -m "first commit"
[main 21e7e4d] first commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 README.md
PS C:\Users\tanve\Desktop\demo project> git remote add origin https://github.com/shrutidarwatkar-sketch/assignment-no-12.git
error: remote origin already exists.
PS C:\Users\tanve\Desktop\demo project> git branch -M main
>> git push -u origin main
remote: Repository not found.
fatal: repository 'https://github.com/YourUsername/Python-Basic-Projects.git/' not found
PS C:\Users\tanve\Desktop\demo project> git remote -v
origin https://github.com/YourUsername/Python-Basic-Projects.git (fetch)
origin https://github.com/YourUsername/Python-Basic-Projects.git (push)
PS C:\Users\tanve\Desktop\demo project> git remote origin
error: unknown subcommand: 'origin'
usage: git remote [-v | --verbose]
or: git remote add [-t <branch>] [-m <master>] [-f] [--tags | --no-tags] [--mirror=<fetch>] <name> <url>
or: git remote rename [--[no-]progress] <old> <new>
or: git remote remove <name>
or: git remote set-head <name> (-a | --auto | -d | --delete | <branch>)
or: git remote [-v | --verbose] show [-n] <name>
or: git remote prune [-n | --dry-run] <name>
or: git remote [-v | --verbose] update [-p | --prune] [(<group> | <remote>)...]
or: git remote set-branches [--add] <name> <branch>...
or: git remote get-url [--push] [--all] <name>
or: git remote set-url [--push] <name> <newurl> [<oldurl>]
or: git remote set-url --add <name> <newurl>
or: git remote set-url --delete <name> <url>

-v, --[no-]verbose    be verbose; must be placed before a subcommand

PS C:\Users\tanve\Desktop\demo project> git remote remove origin
PS C:\Users\tanve\Desktop\demo project> git remote add origin https://github.com/shrutidarwatkar-sketch/assignment-no-12.git
PS C:\Users\tanve\Desktop\demo project> git remote -v
origin https://github.com/shrutidarwatkar-sketch/assignment-no-12.git (fetch)
origin https://github.com/shrutidarwatkar-sketch/assignment-no-12.git (push)
PS C:\Users\tanve\Desktop\demo project> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 14, done.
Counting objects: 100% (14/14), done.
Delta compression using up to 16 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (14/14), 3.83 KiB | 1.52 MiB/s, done.
Total 14 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/shrutidarwatkar-sketch/assignment-no-12.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\tanve\Desktop\demo project>

```

shrutidarvatkar-sketch / assignment-no-12

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SomeName0 first commit 21e7e4d · 11 minutes ago 2 Commits

Q10.py	Add Python assignment no 12	22 minutes ago
Q2.py	Add Python assignment no 12	22 minutes ago
Q3.py	Add Python assignment no 12	22 minutes ago
Q4.py	Add Python assignment no 12	22 minutes ago
Q5.py	Add Python assignment no 12	22 minutes ago
Q6.py	Add Python assignment no 12	22 minutes ago
Q8.py	Add Python assignment no 12	22 minutes ago
Q9.py	Add Python assignment no 12	22 minutes ago
README.md	first commit	11 minutes ago
q7.py	Add Python assignment no 12	22 minutes ago

README

assignment-no-12

About

No description, website, or topics provided.

Readme

Activity

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Create a new release

Packages

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SomeName0

Languages

Python 100.0%

Q2.

PROGRAM:

```
Q2.py > ...
1  def analyze_string(s):
2      if s == "":
3          print("Empty string")
4          return
5
6      print("Length:", len(s))
7
8      print("Reverse:", s[::-1])
9
10     vowels = 0
11     for ch in s.lower():
12         if ch in "aeiou":
13             vowels += 1
14
15     print("Vowels:", vowels)
16
17     print("Characters with indexes:")
18     for i in range(len(s)):
19         print(s[i], "Positive:", i, "Negative:", i-len(s))
20
21
22     text = input("Enter string: ")
23     analyze_string(text)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> & 'C:\Users\tanve\vscode\extensions\ms-python\python\python.exe' 'c:\Users\tanve\vscode\extensions\ms-python\python\python.exe' '53854' '--' 'c:\Users\tanve\Desktop\demo project'
Characters with indexes:
s Positive: 0 Negative: -6
h Positive: 1 Negative: -5
r Positive: 2 Negative: -4
u Positive: 3 Negative: -3
t Positive: 4 Negative: -2
i Positive: 5 Negative: -1
PS C:\Users\tanve\Desktop\demo project> s
```

Q3.

PROGRAM:

```
Q3.py > ...
1  def manage_marks():
2
3      marks = []
4
5      try:
6          for i in range(5):
7              m = float(input("Enter marks: "))
8              marks.append(m)
9
10             print("Marks:", marks)
11             print("Average:", sum(marks)/5)
12             print("Highest:", max(marks))
13             print("Lowest:", min(marks))
14
15             marks.sort(reverse=True)
16             print("Descending order:", marks)
17
18         except ValueError:
19             print("Enter numbers only")
20
21
22  manage_marks()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:\; cd 'c:\ProgramData\Local\Python\pythoncore-3.14-64\python.exe'
2026.6.0-win32-x64\bundled\libs\debugpy\launcher'
.
Enter marks: 87
Enter marks: 90
Enter marks: 89
Enter marks: 78
Enter marks: 67
Marks: [87.0, 90.0, 89.0, 78.0, 67.0]
Average: 82.2
Highest: 90.0
Lowest: 67.0
Descending order: [90.0, 89.0, 87.0, 78.0, 67.0]
PS C:\Users\tanve\Desktop\demo project>
```

Q4.

PROGRAM:

```
Q4.py > ...
1  class Student:
2
3      def __init__(self, name, roll_no):
4          self.name = name
5          self.roll_no = roll_no
6          self.marks_list = []
7
8
9      def add_mark(self, mark):
10         try:
11             if mark < 0:
12                 raise ValueError
13
14             self.marks_list.append(mark)
15
16         except:
17             print("Invalid mark")
18
19
20     def get_average(self):
21         return sum(self.marks_list)/len(self.marks_list)
22
23
24     def display_info(self):
25         print("Name:", self.name)
26         print("Roll No:", self.roll_no)
27         print("Marks:", self.marks_list)
28         print("Average:", self.get_average())
29
30
31 s = Student("ABC",101)
32
33 for i in range(3):
34     try:
35         m=int(input("Enter mark: "))
36         s.add_mark(m)
37     except:
38         print("Invalid input")
39
40
41 s.display_info()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:: c
14-64\python.exe' 'c:\Users\tanve\.vscode\exte
sers\tanve\Desktop\demo project\Q4.py'
Enter mark: 87
Enter mark: 89
Enter mark: 90
Name: ABC
Roll No: 101
Marks: [87, 89, 90]
Average: 88.66666666666667
PS C:\Users\tanve\Desktop\demo project> |
```

Q5.

PROGRAM:

```
Q5.py > ...
1  def student_database():
2
3      students = {}
4
5      while True:
6          print("\n1.Add Student")
7          print("2.Search Student")
8          print("3.Display All")
9          print("4.Exit")
10
11         try:
12             choice = int(input("Enter choice: "))
13
14             if choice == 1:
15                 roll = input("Enter roll number: ")
16                 name = input("Enter name: ")
17                 age = input("Enter age: ")
18                 city = input("Enter city: ")
19
20                 students.update({
21                     roll:{
22                         "name":name,
23                         "age":age,
24                         "city":city
25                     }
26                 })
27                 print("Student Added")
28
29             elif choice == 2:
30                 roll = input("Enter roll number: ")
31                 print(students.get(roll,"Student not found"))
32
33             elif choice == 3:
34                 print(students)
35
36             elif choice == 4:
37                 break
38
39             else:
40                 print("Invalid choice")
41
42         except:
43             print("Invalid input")
44
45     student_database()
```


OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:; cd 'c:\Users\tanve\AppData\Local\Microsoft\WindowsApps\python-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '5864'

1.Add Student
2.Search Student
3.Display All
4.Exit
Enter choice: 1
Enter roll number: 14
Enter name: shruti
Enter age: 18
Enter city: pune
Student Added

1.Add Student
2.Search Student
3.Display All
4.Exit
Enter choice: 2
Enter roll number: 14
{'name': 'shruti', 'age': '18', 'city': 'pune'}

1.Add Student
2.Search Student
3.Display All
4.Exit
Enter choice: 3
{'14': {'name': 'shruti', 'age': '18', 'city': 'pune'}}

1.Add Student
2.Search Student
3.Display All
4.Exit
Enter choice: 4
PS C:\Users\tanve\Desktop\demo project> 
```


Q6.

PROGRAM:

```
Q6.py
1  import random
2  import math
3
4  try:
5
6      numbers = set()
7
8      for i in range(10):
9          n = int(input("Enter number: "))
10         numbers.add(n)
11
12         print("Set:", numbers)
13
14         tup = tuple(numbers)
15
16         print("Tuple:", tup)
17
18         print("Random numbers:",
19               random.sample(tup,3))
20
21         print("Square root:",
22               math.sqrt(sum(tup)))
23
24 except Exception as e:
25     print("Error:",e)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:\python\python32\python.exe c:\python\python32\python.exe
py-2026.6.0-win32-x64\bundled\libs\debugpy\
Enter number: 1
Enter number: 2
Enter number: 3
Enter number: 4
Enter number: 5
Enter number: 6
Enter number: 7
Enter number: 8
Enter number: 9
Enter number: 10
Set: {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}
Tuple: (1, 2, 3, 4, 5, 6, 7, 8, 9, 10)
Random numbers: [5, 7, 4]
Square root: 7.416198487095663
PS C:\Users\tanve\Desktop\demo project>
```

Q7.

PROGRAM:

```
q7.py > ...
1  try:
2
3      square = lambda x:x*x
4
5      numbers = range(1,21)
6
7      squares = []
8
9      for i in numbers:
10         squares.append(square(i))
11
12         print("Squares:",squares)
13
14         print("Even squares:")
15
16         for i in squares:
17             if i % 2 == 0:
18                 print(i)
19
20 except:
21     print("Error")
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:: cd 'c:\Users\tanve\Desktop\demo project'; & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.10_q7.py-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '64767' '--' 'c:\Users\tanve\Desktop\demo project\q7.py'
Squares: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400]
Even squares:
4
16
36
64
100
144
196
256
324
400
Focus folder in explorer (ctrl + click)
PS C:\Users\tanve\Desktop\demo project>
```

Q8.

PROGRAM:

```
Q8.py > ...
1 class Employee:
2
3     def __init__(self, emp_id, name, details):
4         self.emp_id = emp_id
5         self.name = name
6         self.details = details
7
8
9     def show_details(self):
10        print("ID:", self.emp_id)
11        print("Name:", self.name)
12        print("Department:", self.details[0])
13        print("Salary:", self.details[1])
14
15
16 employees = {}
17
18 employees[1] = Employee(1, "Amit", ("IT", 50000))
19 employees[2] = Employee(2, "Rahul", ("HR", 40000))
20 employees[3] = Employee(3, "Sneha", ("Sales", 45000))
21
22
23 for id, emp in employees.items():
24     emp.show_details()
25     print()
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:: cd '
py-2026.6.0-win32-x64\bundled\libs\debugpy\launch
ID: 1
Name: Amit
Department: IT
Salary: 50000

ID: 2
Name: Rahul
Department: HR
Salary: 40000

ID: 3
Name: Sneha
Department: Sales
Sal
-----
Focus folder in explorer (ctrl + click)
PS C:\Users\tanve\Desktop\demo project> 
```

Q9.

PROGRAM:

```
Q9.py > ...
1  import math
2
3  try:
4
5      sentence = input("Enter sentence: ")
6
7      words = set(sentence.split())
8
9      print("Unique words:")
10
11     for w in sorted(words):
12         print(w)
13
14     total = len(words)
15
16     print("Power:",
17         math.pow(total,2))
18
19 except Exception as e:
20     print("Error:",e)
```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> c:: cd 'c:\
py-2026.6.0-win32-x64\bundled\libs\debugpy\launcher
Enter sentence: shruti dnyaneshwar darwatkar
Unique words:
darwatkar
dnyaneshwar
shruti
Power: 9.0
PS C:\Users\tanve\Desktop\demo project> |
```

Q10.

PROGRAM:

```
Q10.py > ...
1  import math
2  import random
3  from datetime import datetime
4
5
6  history = {}
7
8
9  def arithmetic():
10
11      try:
12          a=float(input("Enter first number: "))
13          b=float(input("Enter second number: "))
14
15          print("1.Add")
16          print("2.Sub")
17          print("3.Multiply")
18          print("4.Divide")
19
20          ch=int(input("Choice: "))
21
22          if ch==1:
23              result=a+b
24
25          elif ch==2:
26              result=a-b
27
28          elif ch==3:
29              result=a*b
30
31          elif ch==4:
32              result=a/b
33
34          else:
35              result="Invalid"
36
37          print(result)
38
39          history[str(datetime.now())]=result
40
```

```

40
41     except:
42         print("Error")
43
44
45 while True:
46
47     print("\n1.Basic Arithmetic")
48     print("2.Scientific Calculation")
49     print("3.Random Numbers")
50     print("4.View History")
51     print("5.Exit")
52
53
54     choice=int(input("Enter choice: "))
55
56
57     if choice==1:
58         arithmetic()
59
60
61     elif choice==2:
62
63         n=float(input("Enter number: "))
64
65         print("Square root:",
66             math.sqrt(n))
67
68
69     elif choice==3:
70
71         print(random.randint(1,100))
72
73
74     elif choice==4:
75
76         print(history)
77
78
79     elif choice==5:
80         break
81
82
83     else:
84         print("Invalid choice")

```

OUTPUT:

```
PS C:\Users\tanve\Desktop\demo project> & 'C:\Users\tanve\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.10.0.0_q-b9jbe7qwr' '65211' '--' 'c:\Users\tanve\Desktop\demo project\Q10.py'

1.Basic Arithmetic
2.Scientific Calculation
3.Random Numbers
4.View History
5.Exit
Enter choice: 1
Enter first number: 19
Enter second number: 5
1.Add
2.Sub
3.Multiply
4.Divide
Choice: 1
24.0

1.Basic Arithmetic
2.Scientific Calculation
3.Random Numbers
4.View History
5.Exit
Enter choice: 2
Enter number: 29
Square root: 5.385164807134504

1.Basic Arithmetic
2.Scientific Calculation
3.Random Numbers
4.View History
5.Exit
Enter choice: 6
Invalid choice

1.Basic Arithmetic
2.Scientific Calculation
3.Random Numbers
4.View History
5.Exit
Enter choice: 4
{'2026-06-28 11:45:46.381406': 24.0}

1.Basic Arithmetic
2.Scientific Calculation
3.Random Numbers
4.View History
5.Exit
Enter choice: 5
PS C:\Users\tanve\Desktop\demo project>
```